



On Multipliers of Dirichlet Type Spaces

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Received: 1 July 2014 / Accepted: 16 January 2015 / Published online: 12 February 2015
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Abstract Let $M(\mathcal{D}_K)$ be the class of multipliers of a Dirichlet type space $\mathcal{D}_K$. Under conditions of the weight function K , we first characterize the interpolating sequences for $M(\mathcal{D}_K)$. Secondly, we study boundary values space $\mathcal{D}_K(\partial\mathbb{D})$ and its multiplier spaces $M(\mathcal{D}_K(\partial\mathbb{D}))$. We give a solution of $\bar{\partial}$ -equation in $\mathcal{D}_K(\partial\mathbb{D})$. In addition, by a derivative-free characterization of $\mathcal{D}_K$ spaces and standard techniques, we establish the $\bar{\partial}$ -equation and corona theorem for $M(\mathcal{D}_K)$. Corona type decomposition theorem on $M(\mathcal{D}_K)$ is also obtained.

Keywords Multipliers · Dirichlet type spaces · Interpolating sequences · $\bar{\partial}$ -Equation · Corona theorem

Mathematics Subject Classification 30H05 · 30E05 · 30H80

Communicated by H. Turgay Kaptanoglu.

This work was supported by NSF of China (Nos. 11371234 and 11171203).

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1 Introduction

Let $\mathbb{D}$ and $\partial\mathbb{D}$ be the unit disk and the unit circle in the complex plane $\mathbb{C}$. Denote by $H(\mathbb{D})$ the class of functions analytic in $\mathbb{D}$. As usual, H^∞ is the set of bounded analytic functions in $\mathbb{D}$.

In recent 50 years, interpolating sequences, corona problem and $\bar{\partial}$ -equation of analytic function spaces have been studied extensively (see [3, 4, 7, 14, 19, 20, 23, 35]). These problems have their roots in Carleson's [10, 11] achievement on interpolation and corona theorems for H^∞ .

An important tool used in [10] is that one calls it the Carleson measure nowadays. Given an arc $I \subset \partial\mathbb{D}$, the Carleson box $S(I)$ is defined as

$$S(I) = \{r\zeta \in \mathbb{D} : 1 - |I| < r < 1; \zeta \in I\}.$$

If $|I| \geq 1$, then we set $S(I) = \mathbb{D}$. A positive Borel measure μ on $\mathbb{D}$ is said to be a Carleson measure if

$$\sup_{I \subset \partial\mathbb{D}} \frac{\mu(S(I))}{|I|} < \infty.$$

For $z = x + iy$, define

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right), \quad \frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right).$$

Based on a result due to Carleson [11], Hörmander [19] proved that if $|g(z)|dx dy$ is a Carleson measure, then there exists a function f defined on $\mathbb{D} \cup \partial\mathbb{D}$ such that $\partial f / \partial \bar{z} = g(z)$ on $\mathbb{D}$, and such that the boundary value function f is in $L^\infty(\partial\mathbb{D})$. Later, Jones [20] gave a constructive method to solve the $\bar{\partial}$ -equation.

Motivated by the study above, the goal of this article is to consider the interpolating sequence, corona problem and $\bar{\partial}$ -equation for the class of multipliers of a Dirichlet type space.

Except as specified in this paper, we assume that $K : [0, \infty) \rightarrow [0, \infty)$ is a right-continuous and increasing function. A function $f \in H(\mathbb{D})$ belongs to the Dirichlet type space $\mathcal{D}_K$ if

$$\|f\|_{\mathcal{D}_K}^2 = |f(0)|^2 + \int_{\mathbb{D}} |f'(z)|^2 K(1 - |z|^2) dA(z) < \infty,$$

where $dA(z)$ is the normalized Lebesgue measure on $\mathbb{D}$. Clearly, $\mathcal{D}_K$ is a Hilbert space. In the case $K(t) = t^p$, $0 \leq p < \infty$, the space $\mathcal{D}_K$ gives the usual Dirichlet type space $\mathcal{D}_p$. In particular, if $p = 0$, this gives the classical Dirichlet space $\mathcal{D}$. If $p = 1$, then $\mathcal{D}_p$ is the Hardy space H^2 . We refer to [26, 29, 30] for $\mathcal{D}_p$ spaces. The space $\mathcal{D}_K$ also has been studied extensively. For example, under some conditions on K , Kerman and Sawyer [21] characterized Carleson measures and multipliers of $\mathcal{D}_K$.

in terms of a maximal operator. Aleman [1] proved that each element of the space $\mathcal{D}_K$ can be written as a quotient of two bounded functions in $\mathcal{D}_K$.

Denote by $\mathcal{Q}_K$ the Möbius invariant space generated by $\mathcal{D}_K$. Namely, $\mathcal{Q}_K$ consists of functions $f \in H(\mathbb{D})$ satisfying

$$\sup_{a \in \mathbb{D}} \|f \circ \sigma_a - f(a)\|_{\mathcal{D}_K} < \infty.$$

Here $\sigma_a(z) = (a - z)/(1 - \bar{a}z)$, is the Möbius transformation of $\mathbb{D}$. See [15, 16] for general theory of $\mathcal{Q}_K$ spaces. The following conditions play a crucial role in the study of $\mathcal{Q}_K$ spaces during the last few years:

$$\int_0^1 \frac{\varphi_K(s)}{s} ds < \infty \quad (1.1)$$

and

$$\int_1^\infty \frac{\varphi_K(s)}{s^2} ds < \infty, \quad (1.2)$$

where

$$\varphi_K(s) = \sup_{0 \leq t \leq 1} K(st)/K(t), \quad 0 < s < \infty.$$

Let $M(\mathcal{D}_K)$ be the set of multipliers of $\mathcal{D}_K$, that is,

$$M(\mathcal{D}_K) = \{g \in H(\mathbb{D}) : gf \in \mathcal{D}_K \text{ for all } f \in \mathcal{D}_K\}.$$

In this paper, we study $M(\mathcal{D}_K)$ under the conditions (1.1) and (1.2). We refer to [24] for a survey of multiplier theory on function spaces. We also assume that $K(0) = 0$, otherwise, $\mathcal{D}_K$ is always the Dirichlet space. This paper is organized as follows. In Sect. 2, we give some estimates on the weight function K . In Sect. 3, we show that $\mathcal{D}_K$ has Pick property. Moreover, interpolating sequences for the multiplier space of $\mathcal{D}_K$ are described. In Sect. 4, we study $\mathcal{D}_K(\partial\mathbb{D})$ spaces, the set of boundary values of $\mathcal{D}_K$ functions, in terms of Poisson integral. We characterize the multipliers of $\mathcal{D}_K(\partial\mathbb{D})$ and give a solution of $\bar{\partial}$ -equation in $\mathcal{D}_K(\partial\mathbb{D})$. In Sect. 5, we establish the $\bar{\partial}$ -equation and corona theorem for $M(\mathcal{D}_K)$ spaces.

In addition, the symbol $A \approx B$ means that $A \lesssim B \lesssim A$. We say that $A \lesssim B$ if there exists a constant C such that $A \leq CB$.

2 Weight Function

From [16], we know that if K satisfies (1.1), then

$$K_1(t) = \int_0^t K(s) \frac{ds}{s} \approx K(t), \quad 0 < t < 1.$$

If K satisfies (1.2), then

$$K_2(t) = t \int_t^\infty K(s) \frac{ds}{s^2} \approx K(t), \quad t > 0.$$

Furthermore, there exists a sufficiently small $c > 0$ such that $t^{-c} K_1(t)$ is increasing and $K_2(t)t^{c-1}$ is decreasing. Here we point out that the constant c will appear in this paper frequently. By [34], if K satisfies (1.2), we get that $K(2t) \approx K(t)$ for $0 < t < 1$, K_2 is concave and we can assume that K is differentiable up to any desired order.

First, we prove a result involved the power series expansion of $1/K(1-t)$, $0 \leq t < 1$.

Lemma 2.1 *Suppose that K satisfies (1.1) and (1.2). Then there exist two constants C_1 and C_2 depending only on K such that*

$$C_1 \left(1 + \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n \right) \leq \frac{1}{K(1-t)} \leq C_2 \left(1 + \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n \right)$$

for all $0 \leq t < 1$.

Proof Without loss of generality, we may assume that $1/2 < t < 1$. Hence,

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n &\approx \sum_{n=1}^{\infty} \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{t^{\frac{1}{x}}}{xK(x)} dx \approx \int_0^1 \frac{t^{\frac{1}{x}}}{xK(x)} dx \\ &\approx \int_1^{\infty} \frac{t^y}{yK\left(\frac{1}{y}\right)} dy \approx \int_{-\ln t}^{\infty} \frac{e^{-x}}{xK\left(\frac{1}{x} \ln \frac{1}{t}\right)} dx \\ &\approx \frac{1}{K\left(\ln \frac{1}{t}\right)} \int_{-\ln t}^{\infty} \frac{e^{-x} K\left(\ln \frac{1}{t}\right)}{xK\left(\frac{1}{x} \ln \frac{1}{t}\right)} dx \\ &\lesssim \frac{1}{K\left(\ln \frac{1}{t}\right)} \int_{-\ln t}^{\infty} \frac{e^{-x} \varphi_K(x)}{x} dx. \end{aligned}$$

Since K satisfies (1.1) and (1.2), by [16], there exists a sufficiently small $c > 0$ depending only on K such that

$$\varphi_K(s) \lesssim s^c, \quad 0 < s \leq 1$$

and

$$\varphi_K(s) \lesssim s^{1-c}, \quad s \geq 1.$$

Thus,

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n &\lesssim \frac{1}{K\left(\ln \frac{1}{t}\right)} \int_{-\ln t}^{\infty} \frac{e^{-x} \varphi_K(x)}{x} dx \\ &\lesssim \frac{1}{K\left(\ln \frac{1}{t}\right)} \left(\int_0^{\infty} e^{-x} x^{c-1} dx + \int_0^{\infty} e^{-x} x^{-c} dx \right) \\ &\approx \frac{1}{K(1-t)} (\Gamma(c) + \Gamma(1-c)), \end{aligned}$$

where $\Gamma(\cdot)$ is the Gamma function. It follows that

$$1 + \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n \lesssim \frac{1}{K(1-t)}.$$

On the other hand, since K is increasing, one gets that

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} t^n &\approx \frac{1}{K\left(\ln \frac{1}{t}\right)} \int_{-\ln t}^{\infty} \frac{e^{-x} K\left(\ln \frac{1}{t}\right)}{x K\left(\frac{1}{x} \ln \frac{1}{t}\right)} dx \\ &\gtrsim \frac{1}{K\left(\ln \frac{1}{t}\right)} \int_{\ln 2}^{\infty} \frac{e^{-x} K\left(\ln \frac{1}{t}\right)}{x K\left(\frac{1}{x} \ln \frac{1}{t}\right)} dx \\ &\gtrsim \frac{1}{K(1-t)} \int_{\ln 2}^{\infty} e^{-x} x^{-1} dx \\ &\approx \frac{1}{K(1-t)}. \end{aligned}$$

We obtain the desired result. $\square$

Lemma 2.2 *Let K satisfy (1.1) and (1.2), $0 < \alpha < 2 + c$, $0 < \beta < 2 + c$ and $\alpha + \beta > 3 - c$. Then*

$$\int_{\mathbb{D}} \frac{K(1 - |z|)}{|1 - \bar{\eta}z|^{\alpha} |1 - \bar{\lambda}z|^{\beta}} dA(z) \lesssim \frac{K(|1 - \bar{\eta}\lambda|)}{|1 - \bar{\eta}\lambda|^{\alpha+\beta-2}}$$

holds for all $\eta, \lambda \in \mathbb{D} \cup \partial\mathbb{D}$ except the case $\eta = \lambda \in \partial\mathbb{D}$.

Proof We borrow the idea in [25, 36]. For $\eta, \lambda \in \mathbb{D} \cup \partial\mathbb{D}$ except the case $\eta = \lambda \in \partial\mathbb{D}$, let $d(\eta, \lambda) = |\bar{\eta}(\eta - \lambda)| + |\bar{\lambda}(\lambda - \eta)|$ be the non-isotropic pseudodistance and c_d a constant such that

$$d(\eta, \lambda) \leq c_d (d(\eta, z) + d(z, \lambda)).$$

Following Definition 3.2 in [25], we take a partition of $\mathbb{D}$ as follows,

$$\begin{aligned}\Omega_1 &= \left\{ z \in \mathbb{D} : d(z, \lambda) \leq \frac{d(\eta, \lambda)}{2c_d} \right\}, \quad \Omega_2 = \left\{ z \in \mathbb{D} : d(z, \eta) \leq \frac{d(\eta, \lambda)}{2c_d} \right\}, \\ \Omega_3 &= \left\{ z \in \mathbb{D} : \frac{d(\eta, \lambda)}{2c_d} < d(z, \lambda) \leq d(z, \eta) \right\}, \\ \Omega_4 &= \left\{ z \in \mathbb{D} : \frac{d(\eta, \lambda)}{2c_d} < d(z, \eta) \leq d(z, \lambda) \right\}.\end{aligned}$$

Note that if $\eta = \lambda$, then the both sets Ω_1 and Ω_2 are empty or $\{\eta\}$. By Lemma 3.3 in [25], there exists a constant C such that

$$\begin{aligned}|1 - \bar{z}\lambda| &\leq C|1 - \bar{\eta}\lambda| \leq C|1 - \bar{z}\eta|, \quad z \in \Omega_1, \\ |1 - \bar{z}\eta| &\leq C|1 - \bar{\eta}\lambda| \leq C|1 - \bar{z}\lambda|, \quad z \in \Omega_2, \\ |1 - \bar{\eta}\lambda| &\leq C|1 - \bar{z}\lambda| \leq C|1 - \bar{z}\eta|, \quad z \in \Omega_3, \\ |1 - \bar{\eta}\lambda| &\leq C|1 - \bar{z}\eta| \leq C|1 - \bar{z}\lambda|, \quad z \in \Omega_4.\end{aligned}$$

Hence,

$$\begin{aligned}&\int_{\mathbb{D}} \frac{K(1 - |z|)}{|1 - \bar{\eta}z|^\alpha |1 - \bar{\lambda}z|^\beta} dA(z) \\ &= \int_{\Omega_1 \cup \Omega_3} \frac{K(1 - |z|)}{|1 - \bar{\eta}z|^\alpha |1 - \bar{\lambda}z|^\beta} dA(z) + \int_{\Omega_2 \cup \Omega_4} \frac{K(1 - |z|)}{|1 - \bar{\eta}z|^\alpha |1 - \bar{\lambda}z|^\beta} dA(z) \\ &=: I_1 + I_2.\end{aligned}$$

Since $\alpha > 0$,

$$\begin{aligned}I_1 &= \int_{\Omega_1 \cup \Omega_3} \frac{K(1 - |z|)}{|1 - \bar{\eta}z|^\alpha |1 - \bar{\lambda}z|^\beta} dA(z) \\ &\lesssim \int_{\Omega_1 \cup \Omega_3} \frac{K(1 - |z|)}{(|1 - \bar{\eta}\lambda| + |1 - \bar{z}\lambda|)^\alpha |1 - \bar{\lambda}z|^\beta} dA(z).\end{aligned}$$

Let $\lambda = |\lambda|e^{i\theta}$. Using the change of variable $w = e^{-i\theta}z$ in the last term above, we have

$$I_1 \lesssim \int_{\mathbb{D}} \frac{K(1 - |w|)}{(|1 - \bar{\eta}\lambda| + |1 - |\lambda|w|)^\alpha |1 - |\lambda|w|^\beta} dA(w).$$

Note that

$$|1 - w| \leq 2|1 - |\lambda|w|, \quad \lambda \in \mathbb{D} \cup \partial\mathbb{D}.$$

We get

$$\begin{aligned}
 I_1 &\lesssim \int_{\mathbb{D}} \frac{K(|1-w|)}{(|1-\bar{\eta}\lambda|+|1-w|)^\alpha |1-w|^\beta} dA(w) \\
 &\lesssim \int_0^2 \frac{K(t)}{(|1-\bar{\eta}\lambda|+t)^\alpha t^\beta} t dt \\
 &\lesssim K\left(\frac{|1-\bar{\eta}\lambda|}{2}\right) \int_0^2 \frac{\varphi_K\left(\frac{2t}{|1-\bar{\eta}\lambda|}\right)}{(|1-\bar{\eta}\lambda|+t)^\alpha t^{\beta-1}} dt \\
 &\lesssim \frac{K(|1-\bar{\eta}\lambda|)}{|1-\bar{\eta}\lambda|^{\alpha+\beta-2}} \int_0^{\frac{2}{|1-\bar{\eta}\lambda|}} \frac{\varphi_K(2x)}{(1+x)^\alpha x^{\beta-1}} dx \\
 &\lesssim \frac{K(|1-\bar{\eta}\lambda|)}{|1-\bar{\eta}\lambda|^{\alpha+\beta-2}} \left(\int_0^{\frac{1}{2}} \frac{(2x)^c}{x^{\beta-1}} dx + \int_{\frac{1}{2}}^\infty \frac{(2x)^{1-c}}{x^{\alpha+\beta-1}} dx \right) \\
 &\lesssim \frac{K(|1-\bar{\eta}\lambda|)}{|1-\bar{\eta}\lambda|^{\alpha+\beta-2}}.
 \end{aligned}$$

The last inequality holds since $\beta < 2 + c$ and $\alpha + \beta > 3 - c$. Similarly, we have

$$I_2 \lesssim \frac{K(|1-\bar{\eta}\lambda|)}{|1-\bar{\eta}\lambda|^{\alpha+\beta-2}}.$$

Thus,

$$\int_{\mathbb{D}} \frac{K(1-|z|)}{|1-\bar{\eta}z|^\alpha |1-\bar{\lambda}z|^\beta} dA(z) \lesssim \frac{K(|1-\bar{\eta}\lambda|)}{|1-\bar{\eta}\lambda|^{\alpha+\beta-2}}.$$

The proof is complete. $\square$

Lemma 2.3 *Let K satisfy (1.1) and (1.2). Then*

$$\int_0^1 \frac{r K(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr \lesssim \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2}$$

for all $e^{it}, e^{iu} \in \partial\mathbb{D}$ with $e^{it} \neq e^{iu}$.

Proof Fix $e^{it}, e^{iu} \in \partial\mathbb{D}$ with $e^{it} \neq e^{iu}$. By a technique in [3, page 1296], let N be a positive integer depending on e^{it} and e^{iu} with

$$2^{N-1}|e^{it} - e^{iu}| < 1 < 2^N|e^{it} - e^{iu}|.$$

We write

$$S_0 = \{r \in (0, 1) : 0 < 1-r \leq |e^{it} - e^{iu}|\}$$

and

$$S_n = \{r \in (0, 1) : 2^{n-1}|e^{it} - e^{iu}| < 1 - r \leq 2^n|e^{it} - e^{iu}|\}, \quad n = 1, \dots, N.$$

Note that we can assume that $K(t)/t^c$ is increasing and $K(t)/t$ is decreasing. Since

$$|e^{it} - re^{iu}|^2 = (1 - r)^2 + r|e^{it} - e^{iu}|^2,$$

we deduce that

$$\int_{S_0} \frac{rK(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr \leq \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^{2+c}} \int_{S_0} (1-r)^{c-1} dr \lesssim \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2},$$

and

$$\begin{aligned} \int_{S_n} \frac{rK(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr &\lesssim \int_{S_n} \frac{K(1-r)}{(1-r)^3} dr \\ &\lesssim \frac{K(2^{n-1}|e^{it} - e^{iu}|)}{2^{n-1}|e^{it} - e^{iu}|} \int_{S_n} \frac{1}{(1-r)^2} dr \\ &\approx \frac{K(2^{n-1}|e^{it} - e^{iu}|)}{2^{2n-1}|e^{it} - e^{iu}|^2} \\ &\lesssim \frac{K(|e^{it} - e^{iu}|)}{2^{2n-1}|e^{it} - e^{iu}|^2} \varphi_K(2^{n-1}). \end{aligned}$$

Therefore,

$$\begin{aligned} &\int_0^1 \frac{rK(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr \\ &\leq \int_{S_0} \frac{rK(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr + \sum_{n=1}^N \int_{S_n} \frac{rK(1-r)}{(1-r)|e^{it} - re^{iu}|^2} dr \\ &\lesssim \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2} + \sum_{n=1}^N \frac{K(|e^{it} - e^{iu}|)}{2^{2n-1}|e^{it} - e^{iu}|^2} \varphi_K(2^{n-1}) \\ &\approx \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2} \left(1 + \sum_{n=1}^N \frac{\varphi_K(2^{n-1})}{2^{2n-1}} \right) \\ &\lesssim \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2} \left(1 + \int_1^\infty \frac{\varphi_K(x)}{x^3} dx \right) \\ &\lesssim \frac{K(|e^{it} - e^{iu}|)}{|e^{it} - e^{iu}|^2}. \end{aligned}$$

The proof is finished. □

In this paper, we also need the following estimate of K .

Lemma A [6] *Let K satisfy (1.1) and (1.2). If s and t are any real numbers with $2 - \frac{t}{2} \leq s < 1 + c$, then*

$$\int_{\mathbb{D}} \frac{K(1 - |\sigma_a(w)|^2)}{(1 - |w|^2)^s |1 - \bar{w}z|^t} dA(w) \lesssim \frac{K(1 - |\sigma_a(z)|^2)}{(1 - |z|^2)^{s+t-2}}$$

for all $a, z \in \mathbb{D}$.

3 Pick Property and Interpolating Sequences

In this section, we show that if K satisfies (1.2), then $\mathcal{D}_K$ has the Pick property. Moreover, we characterize the interpolating sequences for the class of multipliers of $\mathcal{D}_K$ spaces.

We assume that $\mathcal{H}$ is a separable Hilbert space of analytic functions in the unit disk. A function g is called a multiplier of $\mathcal{H}$ if $gf \in \mathcal{H}$ whenever $f \in \mathcal{H}$ and we denote the set of multiplier of $\mathcal{H}$ by $M(\mathcal{H})$. Let $R \in H(\mathbb{D})$ and let $\{R_\zeta : \zeta \in \mathbb{D}\}$ be an independent collection of reproducing kernels for $\mathcal{H}$. Here $R_\zeta(z) = R(\bar{\zeta}z)$. The reproducing kernels mean that $f(\zeta) = \langle f, R_\zeta \rangle$ for any $f \in \mathcal{H}$.

We say that $\mathcal{H}$ has the Pick property if whenever $\{z_i\}_{i=1}^\infty \subset \mathbb{D}$ and $\{w_i\}_{i=1}^\infty \subset \mathbb{C}$ such that for each $n \in \mathbb{N}$,

$$\sum_{i,j=1,\dots,n} (1 - \bar{w}_i w_j) \langle R_{z_i}, R_{z_j} \rangle \alpha_i \bar{\alpha}_j \geq 0,$$

for all $\alpha_1, \dots, \alpha_n \in \mathbb{C}$, then there exists $f \in M(\mathcal{H})$ with $\|f\|_{M(\mathcal{H})} \leq 1$ and $f(z_i) = w_i$ for $i = 1, 2, \dots, n$, where

$$\|f\|_{M(\mathcal{H})} = \sup_{\|g\|_{\mathcal{H}}=1} \|fg\|_{\mathcal{H}}$$

and $\|\cdot\|_{\mathcal{H}}$ means the norm of $\mathcal{H}$. See [2, 28] for the Pick property and Hilbert spaces.

Suppose $c_n > 0$ for $n = 0, 1, \dots$. Define an inner product of analytic functions in the unit disk by

$$(f, g) = \sum_{n=0}^{\infty} \frac{1}{c_n} b_n \bar{d}_n,$$

when $f(z) = \sum b_n z^n$ and $g(z) = \sum d_n z^n$. Let

$$R(z) = \sum_{n=0}^{\infty} c_n z^n.$$

Denote by $\mathcal{H}_R$ for the Hilbert space of analytic functions with $\|f\|_{\mathcal{H}_R}^2 \equiv (f, f) < \infty$. Let $R_\zeta(z) = R(\bar{\zeta}z)$, $\zeta \in \mathbb{D}$. Then $R_\zeta(z)$ is the reproducing kernel of $\mathcal{H}_R$ at $\zeta \in \mathbb{D}$, that is, $f(\zeta) = (f, R_\zeta)$ for any $f \in \mathcal{H}_R$.

Using the theory of reproducing kernels and a lemma of Kaluza (see [18, Theorem 22]), Marshall and Sundberg [23] provided the following technique to show that $\mathcal{H}_R$ has the Pick property.

Theorem B Suppose $c_n > 0$ for $n = 0, 1, \dots$, and $c_n^2 \leq c_{n-1}c_{n+1}$ for $n = 1, 2, \dots$. Then $\mathcal{H}_R$ has the Pick property.

Now we return to our primary interest, $\mathcal{D}_K$ spaces.

Lemma 3.1 Let K satisfy (1.2). Define

$$R^K(z) = 1 + \sum_{n=1}^{\infty} \frac{1}{nK\left(\frac{1}{n}\right)} z^n, \quad z \in \mathbb{D}.$$

Then $\mathcal{H}_{R^K} = \mathcal{D}_K$ and $R_{\zeta}^K(z)$ is the reproducing kernel for $\mathcal{H}_{R^K}$ space at $\zeta \in \mathbb{D}$.

Proof Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$. By Theorem 6 of [34], we obtain

$$\|f\|_{\mathcal{D}_K}^2 \approx |a_0|^2 + \sum_{n=1}^{\infty} nK\left(\frac{1}{n}\right) |a_n|^2.$$

By the definition, $f \in \mathcal{H}_{R^K}$ if and only if

$$\|f\|_{\mathcal{H}_{R^K}}^2 = (f, f) = |a_0|^2 + \sum_{n=1}^{\infty} nK\left(\frac{1}{n}\right) |a_n|^2 < \infty.$$

It follows that $\mathcal{H}_{R^K} = \mathcal{D}_K$ and $R_{\zeta}^K(z)$ is the reproducing kernel for $\mathcal{H}_{R^K}$ space at $\zeta \in \mathbb{D}$. $\square$

Recall that

$$K_2(t) = t \int_t^{\infty} K(s) \frac{ds}{s^2} \approx K(t), \quad t > 0.$$

Lemma 3.2 Let K satisfy (1.2). Then $\mathcal{H}_{R^{K_2}}$ has the Pick property.

Proof Since

$$R^{K_2}(z) = 1 + \sum_{n=1}^{\infty} \frac{1}{nK_2\left(\frac{1}{n}\right)} z^n, \quad z \in \mathbb{D},$$

it is easy to see that $nK_2\left(\frac{1}{n}\right) > 0$ for $n = 1, 2, \dots$. Define

$$h(x) = xK_2\left(\frac{1}{x}\right), \quad x > 0.$$

Then

$$h''(x) = \frac{1}{x^3} K_2''\left(\frac{1}{x}\right) = -\frac{1}{x^2} K_2'\left(\frac{1}{x}\right) \leq 0.$$

Thus h is concave which deduces that for $n = 1, 2, \dots$,

$$\frac{h(n+1) + h(n-1)}{2} \leq h(n).$$

Hence

$$h(n+1)h(n-1) \leq h(n)^2,$$

that is,

$$(n+1)K_2\left(\frac{1}{n+1}\right)(n-1)K_2\left(\frac{1}{n-1}\right) \leq n^2 K_2\left(\frac{1}{n}\right)^2.$$

By Theorem B, $\mathcal{H}_{R^{K_2}}$ has the Pick property. $\square$

Let X be a space of analytic functions in the unit disk. A sequence $\{z_n\} \subset \mathbb{D}$ is called an interpolating sequence for X if for each $\{w_n\} \subset \mathbb{C}$, there exists a function $f \in X$ such that $f(z_n) = w_n$ for all $n = 1, 2, \dots$. Note that $\mathcal{D}_K = \mathcal{H}_{R^{K_2}}$ and $\mathcal{H}_{R^{K_2}}$ has the Pick property. The following result is obtained immediately from Corollary 7 in [23].

Theorem 3.3 *Let K satisfy (1.2) and $\{z_n\} \subset \mathbb{D}$. Let $u_n = R_{z_n}^{K_2} / \|R_{z_n}^{K_2}\|_{\mathcal{D}_K}$. Then the following statements are equivalent.*

- (1) $\{z_n\}$ is an interpolating sequence for $M(\mathcal{D}_K)$.
- (2) $f \rightarrow \{f(z_n) / \|R_{z_n}^{K_2}\|_{\mathcal{D}_K}\}$ maps $\mathcal{D}_K$ onto l^2 .
- (3) There exists $C > 0$ such that $\|\sum_n b_n R_{z_n}^{K_2}\|_{\mathcal{D}_K} \leq C \|\sum_n a_n R_{z_n}^{K_2}\|_{\mathcal{D}_K}$ whenever $|b_n| \leq |a_n|$ for all n .
- (4) $\|\sum_n a_n u_n\|^2 \approx \sum_n |a_n|^2$ for all $\{a_n\} \subset l^2$.

A finite positive Borel measure ν on the unit disk $\mathbb{D}$ is called a Carleson measure for the space $\mathcal{H}$ if there exists a constant $C > 0$ such that for every function $f \in \mathcal{H}$,

$$\int_{\mathbb{D}} |f(z)|^2 d\nu(z) \leq C \|f\|_{\mathcal{H}}^2.$$

Next we recall a result in Seip's book [27, page 23].

Theorem C *Let $\mathcal{H}$ be a reproducing kernel Hilbert space of analytic functions on $\mathbb{D}$, and let $Z = \{z_j\} \subset \mathbb{D}$ be a sequence of distinct points. Then (1) $\Rightarrow$ (2) $\Leftrightarrow$ (3) $\Rightarrow$ (4).*

- (1) Z is an interpolating sequence for $M(\mathcal{H})$.

- (2) $\|\sum_j b_j R_{z_j}\|_{\mathcal{H}} \leq \|\sum_j a_j R_{z_j}\|_{\mathcal{H}}$ whenever $|b_j| \leq |a_j|$ for every j .
 (3) $\|\sum_j a_j R_{z_j} / \|R_{z_j}\|_{\mathcal{H}}\|_{\mathcal{H}} \approx \|\{a_j\}\|_{l^2}$.
 (4) Z is $\mathcal{H}$ -separated and $\sum_j \|R_{z_j}\|^{-2} \delta_{z_j}$ is a Carleson measure for $\mathcal{H}$.

Here Z is said to be $\mathcal{H}$ -separated if

$$\sup_{j \neq i} \frac{|R_{z_j}(z_i)|^2}{R_{z_j}(z_j) R_{z_i}(z_i)} < 1.$$

All Hilbert spaces with the Pick property have a property like Theorem 3.3. Next we will show a particular characterization of interpolating sequence of $M(\mathcal{D}_K)$. To prove it, we need the following result due to Kerman and Sawyer [21] (also see [5]).

Theorem D *Let K be increasing, concave and $\lim_{x \rightarrow 0} x/K(x) = 0$. Then, μ is a Carleson measure for $\mathcal{D}_K$ if and only if there exists a constant $C > 0$ such that*

$$\int_I \sup_{\theta \in J \subset I} \frac{\mu(S(J))^2}{K(|J|)|J|} d\theta \leq C\mu(S(I)),$$

for all arcs $I \subset \partial\mathbb{D}$. Here the supremum is taken over all closed arcs $J \subset I$.

Note that if K satisfies (1.2), then the conditions in Theorem D hold for K_2 . A direct corollary of Theorem D is that if μ is a Carleson measure for $\mathcal{D}_K$, then $\mu(S(I)) \leq CK(|I|)$.

Lemma 3.4 *Suppose that K satisfies (1.1) and (1.2). Let*

$$B(z) = \prod_n \frac{|z_n|}{z_n} \frac{z_n - z}{1 - \bar{z}_n z}$$

be the Blaschke product with zeros $\{z_n\} \subset \mathbb{D}$. If $\sum_n K(1 - |z_n|)\delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$, then $B \in M(\mathcal{D}_K)$.

Proof Since K satisfies (1.2), by [16], we can assume that $t/K(t)$ is increasing. For any $I \subseteq \partial\mathbb{D}$,

$$\sum_{z_n \in S(I)} (1 - |z_n|) \lesssim \sum_{z_n \in S(I)} K(1 - |z_n|)|I|/K(|I|).$$

Since $\sum_n K(1 - |z_n|)\delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$, by Theorem D,

$$\sum_{z_n \in S(I)} K(1 - |z_n|) \lesssim K(|I|).$$

It follows that $\sum_n (1 - |z_n|)\delta_{z_n}$ is a Carleson measure. From [17, page 277],

$$\sup_{z \in \mathbb{D}} \sum_n \left(1 - |\sigma_{z_n}(z)|^2\right) < \infty.$$

To show $B \in M(\mathcal{D}_K)$, it suffices to prove that for any $f \in \mathcal{D}_K$, we have

$$\int_{\mathbb{D}} |f(z)|^2 |B'(z)|^2 K(1 - |z|^2) dA(z) < \infty.$$

Note that $(1 - |z|^2)|B'(z)| \lesssim 1$ and

$$|B'(z)| \leq \sum_n \frac{1 - |z_n|^2}{|1 - \bar{z}_n z|^2}.$$

We obtain

$$\begin{aligned} & \int_{\mathbb{D}} |f(z)|^2 |B'(z)|^2 K(1 - |z|^2) dA(z) \\ & \lesssim \int_{\mathbb{D}} |f(z)|^2 \sum_n \frac{1 - |z_n|^2}{|1 - \bar{z}_n z|^2} \frac{K(1 - |z|^2)}{1 - |z|^2} dA(z) \\ & \lesssim \sum_n (1 - |z_n|^2) \int_{\mathbb{D}} \frac{|f(z_n)|^2}{|1 - \bar{z}_n z|^2} \frac{K(1 - |z|^2)}{1 - |z|^2} dA(z) \\ & \quad + \sum_n (1 - |z_n|^2) \int_{\mathbb{D}} \frac{|f(z) - f(z_n)|^2}{|1 - \bar{z}_n z|^2} \frac{K(1 - |z|^2)}{1 - |z|^2} dA(z) \\ & =: I_1 + I_2. \end{aligned}$$

Using Lemma A and the fact that $\sum_n K(1 - |z_n|)\delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$, we obtain

$$I_1 \lesssim \sum_n |f(z_n)|^2 K(1 - |z_n|^2) \lesssim \|f\|_{\mathcal{D}_K}^2.$$

By the change of variable $w = \sigma_{z_n}(z)$, one gets that

$$I_2 \approx \sum_n \int_{\mathbb{D}} |f \circ \sigma_{z_n}(w) - f(z_n)|^2 \frac{K(1 - |\sigma_{z_n}(w)|^2)}{(1 - |w|^2)} dA(w).$$

Following the argument developed in [6, Theorem 2.3], we have

$$I_2 \lesssim \sum_n \int_{\mathbb{D}} |f'(z)|^2 (1 - |\sigma_{z_n}(z)|^2) K(1 - |z|^2) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2.$$

The proof is finished. □

Earl [14] gave a constructive proof of Carleson interpolating theorem for bounded analytic functions. For a sequence $\{z_n\} \subset \mathbb{D}$ with

$$\inf_k \prod_{j \neq k} \rho(z_j, z_k) = \delta > 0, \quad \rho(z_j, z_k) = |\sigma_{z_j}(z_k)|,$$

and a sequence $\{w_n\} \in l^\infty$, Earl constructed a Blaschke product B with zeros $\{\zeta_n\}$ such that for some constant C , the function $f(z) = C\|\{w_n\}\|_{l^\infty} B(z)$ solves the interpolating problem

$$f(z_n) = w_n, \quad n = 1, 2, \dots$$

Furthermore, zeros $\{\zeta_n\}$ of the Blaschke product B satisfy

$$\rho(z_n, \zeta_n) \leq \delta/3.$$

A sequence $\{a_n\} \subset \mathbb{D}$ is called a separated sequence if

$$\inf_{j \neq k} \rho(a_j, a_k) > 0.$$

The following theorem is the main result of this section.

Theorem 3.5 *Let K satisfy (1.1) and (1.2). Then the following conditions are equivalent.*

- (1) $\{z_n\}_{n=1}^\infty$ is an interpolating sequence for $M(\mathcal{D}_K)$.
- (2) $\{z_n\}_{n=1}^\infty$ is separated and $\sum_n K(1 - |z_n|)\delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$.

Proof (1) $\Rightarrow$ (2). By [21], $M(\mathcal{D}_K) \subset H^\infty$. Thus $\{z_n\}$ is also an interpolating sequence for H^∞ . By Carleson's interpolating theorem, we obtain that $\{z_n\}$ is separated. Theorem C gives that $\sum_n \|R_{z_n}^{K_2}\|_{\mathcal{D}_K}^{-2} \delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$. By Lemma 2.1,

$$\|R_{z_n}^{K_2}\|_{\mathcal{D}_K}^2 \approx 1 + \sum_{j=1}^\infty \frac{1}{jK(\frac{1}{j})} |z_n^j|^2 \approx 1/K(1 - |z_n|).$$

Thus $\sum_n K(1 - |z_n|)\delta_{z_n}$ is a Carleson measure for $\mathcal{D}_K$.

(2) $\Rightarrow$ (1). By the proof of Lemma 3.4, $\sum_n (1 - |z_n|)\delta_{z_n}$ is a Carleson measure. Combining this with the separation condition, Carleson's interpolating theorem yields that

$$\inf_k \prod_{j:j \neq k} \rho(z_j, z_k) = \delta > 0.$$

Let $a = \{a_j\}$ be any sequence in l^∞ . By Earl's result, there is a constant C and a Blaschke product B with zeros $\{\zeta_j\}$ such that $f(z) = C\|a\|_{l^\infty} B(z)$ satisfying

$$f(z_j) = a_j, \quad j = 1, 2, \dots$$

By Lemma 3.4, it is sufficient to show that $\sum K(1 - |\zeta_j|)\delta_{\zeta_j}$ is a Carleson measure for $\mathcal{D}_K$.

For any $g \in \mathcal{D}_K$,

$$\begin{aligned} \sum_j |g(\zeta_j)|^2 K(1 - |\zeta_j|^2) &\lesssim \sum_j |g(\zeta_j) - g(z_j)|^2 K(1 - |\zeta_j|^2) \\ &\quad + \sum_j |g(z_j)|^2 K(1 - |\zeta_j|^2) \\ &=: S_1 + S_2. \end{aligned}$$

The condition $\rho(z_j, \zeta_j) < \delta/3 < 1$ deduces that

$$1 - |\zeta_j|^2 \approx 1 - |z_j|^2.$$

Since $\sum_j K(1 - |z_j|^2)\delta_{z_j}$ is a Carleson measure for $\mathcal{D}_K$, we obtain

$$S_2 \lesssim \sum_j |g(z_j)|^2 K(1 - |z_j|^2) \lesssim \|g\|_{\mathcal{D}_K}^2.$$

Denote by $[z_j, \zeta_j]$ the line segment jointing z_j and ζ_j . The subharmonicity of $|g'|$ gives

$$\begin{aligned} |g(\zeta_j) - g(z_j)| &\leq \int_{[z_j, \zeta_j]} |g'(w)| |dw| \\ &\lesssim \int_{[z_j, \zeta_j]} \left(\frac{1}{(1 - |w|)^2} \int_{D_w} |g'(z)| dA(z) \right) |dw|, \end{aligned}$$

where $D_w = \{z: \rho(z, w) < \delta/6\}$. Since $\rho(z_j, \zeta_j) < \delta/3$, we have $D_w \subset D_j$ for each $w \in [z_j, \zeta_j]$, where $D_j = \{z: \rho(z, z_j) < \delta/2\}$. Note that $1 - |w| \approx 1 - |z_j|$. We get

$$|g(\zeta_j) - g(z_j)| \lesssim \frac{|z_j - \zeta_j|}{(1 - |z_j|)^2} \int_{D_j} |g'(z)| dA(z).$$

Note that the pseudohyperbolic disks D_j are pairwise disjoint and $|z_j - \zeta_j| \lesssim |1 - z_j \overline{\zeta_j}| \approx 1 - |z_j|$. By Hölder's inequality,

$$\begin{aligned} S_1 &\lesssim \sum_j \left(\int_{D_j} |g'(z)| dA(z) \right)^2 \frac{K(1 - |z_j|^2)}{(1 - |z_j|^2)^2} \\ &\lesssim \sum_j \int_{D_j} |g'(z)|^2 dA(z) |D_j| \frac{K(1 - |z_j|^2)}{(1 - |z_j|^2)^2} \\ &\lesssim \sum_j \int_{D_j} |g'(z)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim \|g\|_{\mathcal{D}_K}^2. \end{aligned}$$

Therefore, $\sum K(1 - |\zeta_j|)\delta_{\zeta_j}$ is a Carleson measure for $\mathcal{D}_K$. The proof is complete. $\square$

- Remark 3.6* (1) Interpolating sequences for $M(\mathcal{D})$ were studied by Marshall and Sundberg [23] and Bishop [7] simultaneously in 1994 with different techniques. Xiao [35] characterized interpolating sequences of $M(\mathcal{D}_p)$, $0 < p < 1$, by applying Jones's [20] method. Interpolating sequences for multipliers of Besov type spaces were studied in [3] by Eral's constructive proof of Carleson interpolating theorem. The technique in the proof of Theorem 3.5 is from [3, 23].
- (2) Marshall and Sundberg [23] showed that if a Hilbert space $\mathcal{H}$ has the Pick property, then the interpolating sequences for $\mathcal{H}$ and for $M(\mathcal{H})$ are the same. If $\rho(z) = K(1 - |z|)$ in [8], Theorem 3.5 gives an answer to Problem 2 in [8].

4 Boundary Values

If K satisfies (1.2), then $\mathcal{D}_K$ is a subset of the Hardy space H^2 . See [13, 37] for Hardy spaces. Thus any function f in $\mathcal{D}_K$ has non-tangential limits a.e. on $\partial\mathbb{D}$. In this section, we first characterize $\mathcal{D}_K(\partial\mathbb{D})$ spaces, the set of boundary values of $\mathcal{D}_K$ functions, in terms of Poisson integral. Secondly, we describe the multipliers of $\mathcal{D}_K(\partial\mathbb{D})$ spaces. At last, we solve $\bar{\partial}$ -equation in $\mathcal{D}_K(\partial\mathbb{D})$ spaces.

Denote by $\mathcal{D}_K(\partial\mathbb{D})$ the space of functions $f \in L^2(\partial\mathbb{D})$ with

$$|||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 = \int_0^{2\pi} \int_0^{2\pi} \frac{|f(e^{it}) - f(e^{is})|^2}{|e^{it} - e^{is}|^2} K(|e^{it} - e^{is}|) ds dt < \infty.$$

The norm for $\mathcal{D}_K(\partial\mathbb{D})$ is defined as

$$\|f\|_{\mathcal{D}_K(\partial\mathbb{D})} = \left(\|f\|_{L^2(\partial\mathbb{D})}^2 + |||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 \right)^{1/2}.$$

Let

$$\widehat{f}(z) = \int_0^{2\pi} f(e^{it}) \frac{1 - |z|^2}{|e^{it} - z|^2} \frac{dt}{2\pi}$$

be the Poisson integral of f .

We recall the following Hardy-type inequalities in [22].

Lemma E Let $0 < b \leq \infty$, $1 < p < \infty$ and $p' = \frac{p}{p-1}$. Assumed that functions μ and v are measurable and non-negative in the interval $(0, b)$. Then

$$\int_0^b \left(\int_0^s f(t) dt \right)^p \mu(s) ds \leq C \int_0^b f(s)^p v(s) ds$$

holds for all measurable functions $f \geq 0$ if and only if

$$A =: \sup_{0 < s < b} \left(\int_s^b \mu(t) dt \right)^{\frac{1}{p}} \left(\int_0^s v(t)^{1-p'} dt \right)^{\frac{1}{p'}} < \infty.$$

And

$$\int_0^b \left(\int_s^b f(t) dt \right)^p \mu(s) ds \leq C \int_0^b f(s)^p \nu(s) ds$$

holds for all measurable functions $f \geq 0$ if and only if

$$B =: \sup_{0 < s < b} \left(\int_0^s \mu(t) dt \right)^{\frac{1}{p}} \left(\int_s^b \nu(t)^{1-p'} dt \right)^{\frac{1}{p'}} < \infty.$$

Here C depends only on p , A or B .

Let $f \in C^1(\mathbb{D})$, the space of functions on $\mathbb{D}$ with continuous partial derivatives. The gradient of f is

$$\nabla f(z) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

First, we give a characterization of $\mathcal{D}_K(\partial\mathbb{D})$ by Poisson integral.

Theorem 4.1 *Let K satisfy (1.1) and (1.2) and $f \in L^2(\partial\mathbb{D})$. Then $f \in \mathcal{D}_K(\partial\mathbb{D})$ if and only if*

$$\int_{\mathbb{D}} |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z) < \infty.$$

Proof Necessity. By the definition of Poisson integral and a computation, we have

$$|\nabla \widehat{f}(re^{iu})| \lesssim \int_0^{2\pi} \frac{|f(e^{it}) - f(e^{iu})|}{|e^{it} - re^{iu}|^2} dt.$$

From Hölder's inequality,

$$\begin{aligned} |\nabla \widehat{f}(re^{iu})|^2 &\lesssim \left(\int_0^{2\pi} \frac{|f(e^{it}) - f(e^{iu})|}{|e^{it} - re^{iu}|^2} dt \right)^2 \\ &\lesssim \int_0^{2\pi} \frac{|f(e^{it}) - f(e^{iu})|^2}{|e^{it} - re^{iu}|^2} dt \int_0^{2\pi} \frac{1}{|e^{it} - re^{iu}|^2} dt \\ &\lesssim (1 - r^2)^{-1} \int_0^{2\pi} \frac{|f(e^{it}) - f(e^{iu})|^2}{|e^{it} - re^{iu}|^2} dt. \end{aligned}$$

Using Lemma 2.3, one gets that

$$\begin{aligned} &\int_{\mathbb{D}} |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim \int_0^{2\pi} \int_0^{2\pi} |f(e^{it}) - f(e^{iu})|^2 \int_0^1 \frac{rK(1 - r^2)}{(1 - r^2)|e^{it} - re^{iu}|^2} dr dt du \end{aligned}$$

$$\begin{aligned} &\lesssim \int_0^{2\pi} \int_0^{2\pi} \frac{|f(e^{it}) - f(e^{iu})|^2}{|e^{it} - e^{iu}|^2} K(|e^{it} - e^{iu}|) dt du \\ &\approx |||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2. \end{aligned}$$

Sufficiency. Note that $|e^{it} - 1| \approx t$ for $t \in (0, \pi)$,

$$\begin{aligned} |||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 &= \int_{-\pi}^{\pi} \frac{K(|e^{ih} - 1|)}{|e^{ih} - 1|^2} \int_0^{2\pi} |f(e^{i(u+h)}) - f(e^{iu})|^2 dudh \\ &\approx \int_0^{\pi} \frac{K(h)}{h^2} \int_0^{2\pi} |f(e^{i(u+h)}) - f(e^{iu})|^2 dudh, \end{aligned}$$

and

$$\int_{1/2}^{\pi} \frac{K(h)}{h^2} \int_0^{2\pi} |f(e^{i(u+h)}) - f(e^{iu})|^2 dudh \lesssim \int_0^{2\pi} |f(e^{iu})|^2 du \approx \|f\|_{L^2(\partial\mathbb{D})}^2,$$

we get

$$|||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 \lesssim \int_0^{1/2} \frac{K(h)}{h^2} \int_0^{2\pi} |f(e^{i(u+h)}) - f(e^{iu})|^2 dudh + \|f\|_{L^2(\partial\mathbb{D})}^2.$$

Let $r = 1 - h$.

$$\begin{aligned} &|f(e^{i(u+h)}) - f(e^{iu})| \\ &\leq |f(e^{i(u+h)}) - \widehat{f}(re^{i(u+h)})| + |\widehat{f}(re^{i(u+h)}) - \widehat{f}(re^{iu})| + |\widehat{f}(re^{iu}) - f(e^{iu})| \\ &\leq \int_r^1 |\nabla \widehat{f}(xe^{i(u+h)})| dx + \int_0^h |\nabla \widehat{f}(re^{i(u+t)})| dt + \int_r^1 |\nabla \widehat{f}(xe^{iu})| dx. \end{aligned}$$

Applying Minkowski's inequality gives

$$\begin{aligned} &\int_0^{2\pi} |f(e^{i(u+h)}) - f(e^{iu})|^2 du \\ &\lesssim \int_0^{2\pi} \left(\int_r^1 |\nabla \widehat{f}(xe^{iu})| dx \right)^2 du + \int_0^{2\pi} \left(\int_0^h |\nabla \widehat{f}(re^{i(u+t)})| dt \right)^2 du \\ &\lesssim \left(\int_r^1 \left(\int_0^{2\pi} |\nabla \widehat{f}(xe^{iu})|^2 du \right)^{\frac{1}{2}} dx \right)^2 \\ &\quad + \left(\int_0^h \left(\int_0^{2\pi} |\nabla \widehat{f}(re^{i(u+t)})|^2 du \right)^{\frac{1}{2}} dt \right)^2. \end{aligned}$$

Note that

$$\sup_{0 < s < 1/2} \int_s^{1/2} \frac{K(t)}{t^2} dt \int_0^s \frac{1}{K(t)} dt \lesssim \sup_{0 < s < 1/2} \frac{K(s)}{s^{1-c}} \int_s^{1/2} \frac{1}{t^{1+c}} dt \frac{s^{1-c}}{K(s)} \int_0^s t^{c-1} dt \lesssim c^{-2}.$$

By the change of variable $x = 1 - t$, and Lemma E, we obtain

$$\begin{aligned} & \int_0^{1/2} \frac{K(h)}{h^2} \left(\int_r^1 \left(\int_0^{2\pi} |\nabla \widehat{f}(xe^{iu})|^2 du \right)^{\frac{1}{2}} dx \right)^2 dh \\ &= \int_0^{1/2} \frac{K(h)}{h^2} \left(\int_0^h \left(\int_0^{2\pi} |\nabla \widehat{f}((1-t)e^{iu})|^2 du \right)^{\frac{1}{2}} dt \right)^2 dh \\ &\lesssim \int_0^{1/2} \int_0^{2\pi} |\nabla \widehat{f}((1-h)e^{iu})|^2 K(h) du dh \\ &\lesssim \int_{\mathbb{D}} |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z). \end{aligned}$$

On the other hand,

$$\begin{aligned} & \int_0^{1/2} \frac{K(h)}{h^2} \left(\int_0^h \left(\int_0^{2\pi} |\nabla \widehat{f}(re^{i(u+t)})|^2 du \right)^{\frac{1}{2}} dt \right)^2 dh \\ &= \int_0^{1/2} \frac{K(h)}{h^2} \left(\int_0^h \left(\int_0^{2\pi} |\nabla \widehat{f}(re^{iu})|^2 du \right)^{\frac{1}{2}} dt \right)^2 dh \\ &= \int_0^{1/2} K(h) \left(\int_0^{2\pi} |\nabla \widehat{f}((1-h)e^{iu})|^2 du \right) dh \\ &\lesssim \int_{\mathbb{D}} |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z). \end{aligned}$$

Therefore,

$$|||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 \lesssim \int_{\mathbb{D}} |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z) + \|f\|_{L^2(\partial\mathbb{D})}^2.$$

The theorem is proved. $\square$

Remark 4.2 Let $f \in L^2(\partial\mathbb{D})$ and $F \in C^1(\mathbb{D})$ with $\lim_{r \rightarrow 1} F(re^{it}) = f(e^{it})$ a.e. $e^{it} \in \partial\mathbb{D}$. The proof of Theorem 4.1 implies

$$|||f|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 \lesssim \int_{\mathbb{D}} |\nabla F(z)|^2 K(1 - |z|^2) dA(z) + \|f\|_{L^2(\partial\mathbb{D})}^2.$$

Corollary 4.3 *Let K satisfy (1.1) and (1.2), and $f \in H^2$. Then $f \in \mathcal{D}_K$ if and only if $f_b \in \mathcal{D}_K(\partial\mathbb{D})$, where f_b denotes the boundary values of f .*

Proof If $f \in H^2$, then $f = \widehat{f}_b$ (see [17]). The desired result is a consequence of Theorem 4.1. $\square$

Next we describe the Carleson measure for $\mathcal{D}_K$ using functions in $\mathcal{D}_K(\partial\mathbb{D})$.

Lemma 4.4 *Let K satisfy (1.1) and (1.2), and let μ be a positive Borel measure on $\mathbb{D}$. Then μ is a Carleson measure for $\mathcal{D}_K$ if and only if*

$$\int_{\mathbb{D}} |\widehat{h}(z)|^2 d\mu(z) \lesssim \|h\|_{\mathcal{D}_K(\partial\mathbb{D})}^2$$

for all $h \in \mathcal{D}_K(\partial\mathbb{D})$.

Proof Necessity. Let $h \in \mathcal{D}_K(\partial\mathbb{D})$. Without loss of generality, we may assume that h is real. Denote by $\widetilde{h}$ the harmonic conjugate function of $\widehat{h}$. Using the Cauchy–Riemann equation, we have $|\nabla\widehat{h}(z)| \approx |f'(z)|$, where $f = \widehat{h} + i\widetilde{h}$. Theorem 4.1 implies $f \in \mathcal{D}_K$. Since μ is a Carleson measure for $\mathcal{D}_K$,

$$\begin{aligned} \int_{\mathbb{D}} |\widehat{h}(z)|^2 d\mu(z) &\lesssim \int_{\mathbb{D}} |f(z)|^2 d\mu(z) \lesssim \|f\|_{\mathcal{D}_K}^2 \\ &\approx |f(0)|^2 + \int_{\mathbb{D}} |\nabla\widehat{h}(z)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim \|h\|_{\mathcal{D}_K(\partial\mathbb{D})}^2. \end{aligned}$$

Sufficiency. Let $f \in \mathcal{D}_K$. By Corollary 4.3, $f_b \in \mathcal{D}_K(\partial\mathbb{D})$ and $f = \widehat{f}_b$. Thus,

$$\int_{\mathbb{D}} |f(z)|^2 d\mu(z) = \int_{\mathbb{D}} |\widehat{f}_b(z)|^2 d\mu(z) \lesssim \|f_b\|_{\mathcal{D}_K(\partial\mathbb{D})}^2 \lesssim \|f\|_{\mathcal{D}_K}^2.$$

The proof is complete. $\square$

Lemma 4.5 *Let K satisfy (1.1) and (1.2). Then*

$$\int_{\mathbb{D}} |\widehat{f}|^2(z) |\nabla\widehat{g}(z)|^2 K(1 - |z|^2) dA(z) \lesssim \int_{\partial\mathbb{D}} |f(\lambda)|^2 \omega_\lambda^2(g) |d\lambda|,$$

for all $f, g \in L^2(\partial\mathbb{D})$. Here

$$\omega_\lambda^2(g) = \int_0^{2\pi} \frac{|g(\lambda) - g(e^{iu})|^2}{|\lambda - e^{iu}|^2} K(|\lambda - e^{iu}|) du.$$

Proof By a similar calculation as in the proof of Theorem 4.1, we obtain that

$$|\nabla\widehat{g}(z)|^2 \lesssim (1 - |z|^2)^{-1} \int_0^{2\pi} \frac{|g(e^{it}) - g(\lambda)|^2}{|e^{it} - z|^2} dt$$

for any $\lambda \in \partial\mathbb{D}$. Applying Poisson integral formula, Fubini's theorem and Lemma 2.2, one gets

$$\begin{aligned}
 & \int_{\mathbb{D}} |\widehat{f}|^2(z) |\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z) \\
 & \approx \int_{\partial\mathbb{D}} |f(\lambda)|^2 \int_{\mathbb{D}} |\nabla \widehat{g}(z)|^2 \frac{(1 - |z|^2)}{|1 - \bar{\lambda}z|^2} K(1 - |z|^2) dA(z) |d\lambda| \\
 & \lesssim \int_{\partial\mathbb{D}} |f(\lambda)|^2 |d\lambda| \int_0^{2\pi} |g(e^{it}) - g(\lambda)|^2 \int_{\mathbb{D}} \frac{K(1 - |z|^2)}{|1 - \bar{\lambda}z|^2 |1 - e^{-it}z|^2} dA(z) dt \\
 & \lesssim \int_{\partial\mathbb{D}} |f(\lambda)|^2 \int_0^{2\pi} |g(e^{it}) - g(\lambda)|^2 \frac{K(|e^{it} - \lambda|)}{|e^{it} - \lambda|^2} |d\lambda| dt \\
 & \approx \int_{\partial\mathbb{D}} |f(\lambda)|^2 \omega_{\lambda}^2(g) |d\lambda|.
 \end{aligned}$$

□

Let $M(\mathcal{D}_K(\partial\mathbb{D}))$ be the class of multipliers of $\mathcal{D}_K(\partial\mathbb{D})$, that is,

$$M(\mathcal{D}_K(\partial\mathbb{D})) = \{g \in \mathcal{D}_K(\partial\mathbb{D}) : fg \in \mathcal{D}_K(\partial\mathbb{D}) \text{ whenever } f \in \mathcal{D}_K(\partial\mathbb{D})\}.$$

As one of main results of this section, we next give a characterization of $M(\mathcal{D}_K(\partial\mathbb{D}))$.

Theorem 4.6 *Let K satisfy (1.1) and (1.2). Then $g \in M(\mathcal{D}_K(\partial\mathbb{D}))$ if and only if $g \in L^\infty(\partial\mathbb{D})$ and $|\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$.*

Proof Sufficiency. Let $f \in \mathcal{D}_K(\partial\mathbb{D})$. Since $g \in L^\infty(\partial\mathbb{D})$, it follows that $fg \in L^2(\partial\mathbb{D})$. By Remark 4.2, we have

$$\begin{aligned}
 |||fg|||_{\mathcal{D}_K(\partial\mathbb{D})}^2 & \lesssim \int_{\mathbb{D}} |\nabla(\widehat{f}\widehat{g})(z)|^2 K(1 - |z|^2) dA(z) + \|fg\|_{L^2(\partial\mathbb{D})}^2 \\
 & \lesssim \int_{\mathbb{D}} |\widehat{g}(z)|^2 |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z) \\
 & \quad + \int_{\mathbb{D}} |\widehat{f}(z)|^2 |\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z) + \|fg\|_{L^2(\partial\mathbb{D})}^2.
 \end{aligned}$$

Note that $\widehat{g} \in L^\infty(\mathbb{D})$. Using Theorem 4.1, we get

$$\int_{\mathbb{D}} |\widehat{g}(z)|^2 |\nabla \widehat{f}(z)|^2 K(1 - |z|^2) dA(z) \lesssim \|\widehat{g}\|_{L^\infty}^2 \|f\|_{\mathcal{D}_K(\partial\mathbb{D})}^2.$$

On the other hand, since $|\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, by Lemma 4.4, we obtain

$$\int_{\mathbb{D}} |\widehat{f}(z)|^2 |\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z) \lesssim \|f\|_{\mathcal{D}_K(\partial\mathbb{D})}^2.$$

Thus, $g \in M(\mathcal{D}_K(\partial\mathbb{D}))$.

Necessity. Let $g \in M(\mathcal{D}_K(\partial\mathbb{D}))$. First, we use the technique in [12] to prove $g \in L^\infty(\partial\mathbb{D})$. By the Closed Graph Theorem, there exists a constant C_g depending only on g such that

$$\|gf\|_{\mathcal{D}_K(\partial\mathbb{D})} \leq C_g \|f\|_{\mathcal{D}_K(\partial\mathbb{D})}$$

for all $f \in \mathcal{D}_K(\partial\mathbb{D})$. Taking $\Phi = g/C_g$, we get that

$$\|\Phi f\|_{\mathcal{D}_K(\partial\mathbb{D})} \leq \|f\|_{\mathcal{D}_K(\partial\mathbb{D})}$$

for all $f \in \mathcal{D}_K(\partial\mathbb{D})$. Since $1 \in \mathcal{D}_K(\partial\mathbb{D})$, we obtain that $\Phi^n \in \mathcal{D}_K(\partial\mathbb{D})$ with $\|\Phi^n\|_{\mathcal{D}_K(\partial\mathbb{D})} \leq \|\Phi\|_{\mathcal{D}_K(\partial\mathbb{D})}$ for $n \geq 1$. Write

$$\Phi^n(e^{it})(f(e^{it}) - f(e^{iu})) = (\Phi^n f)(e^{it}) - (\Phi^n f)(e^{iu}) - f(e^{iu})(\Phi^n(e^{it}) - \Phi^n(e^{iu})).$$

Hence,

$$\begin{aligned} & \int_0^{2\pi} \int_0^{2\pi} |\Phi(e^{it})|^{2n} \frac{|f(e^{it}) - f(e^{iu})|^2}{|e^{it} - e^{iu}|^2} K(|e^{it} - e^{iu}|) dt du \\ & \leq 2\|\Phi^n f\|_{\mathcal{D}_K(\partial\mathbb{D})}^2 + 2 \int_0^{2\pi} \int_0^{2\pi} |f(e^{iu})|^2 \frac{|\Phi^n(e^{it}) - \Phi^n(e^{iu})|^2}{|e^{it} - e^{iu}|^2} K(|e^{it} - e^{iu}|) dt du \\ & \leq 2\|f\|_{\mathcal{D}_K(\partial\mathbb{D})}^2 + 2 \int_0^{2\pi} \int_0^{2\pi} |f(e^{iu})|^2 \frac{|\Phi^n(e^{it}) - \Phi^n(e^{iu})|^2}{|e^{it} - e^{iu}|^2} K(|e^{it} - e^{iu}|) dt du. \end{aligned}$$

Let f be the identity function on $\partial\mathbb{D}$ in the above inequality. It follows that

$$\begin{aligned} & \int_0^{2\pi} \int_0^{2\pi} |\Phi(e^{it})|^{2n} K(|e^{it} - e^{iu}|) dt du \\ & \lesssim 1 + \int_0^{2\pi} \int_0^{2\pi} \frac{|\Phi^n(e^{it}) - \Phi^n(e^{iu})|^2}{|e^{it} - e^{iu}|^2} K(|e^{it} - e^{iu}|) dt du \\ & \lesssim 1 + \|\Phi\|_{\mathcal{D}_K(\partial\mathbb{D})}^2. \end{aligned}$$

Since n is arbitrary, the inequality above gives $g = C_g \Phi \in L^\infty(\partial\mathbb{D})$.

Next, we show that $|\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. For $f \in \mathcal{D}_K$, write

$$f(\lambda)(g(e^{it}) - g(\lambda)) = (fg)(e^{it}) - (fg)(\lambda) + g(e^{it})(f(\lambda) - f(e^{it})).$$

We have

$$|f(\lambda)|^2 \omega_\lambda^2(g) \lesssim \omega_\lambda^2(fg) + \|g\|_\infty^2 \omega_\lambda^2(f).$$

From $|f|^2 \leq \widehat{|f|^2}$ and Lemma 4.5, we deduce

$$\begin{aligned} \int_{\mathbb{D}} |f(z)|^2 |\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z) &\lesssim \int_{\partial \mathbb{D}} |f(\lambda)|^2 \omega_\lambda^2(g) |d\lambda| \\ &\lesssim \int_{\partial \mathbb{D}} \omega_\lambda^2(fg) |d\lambda| + \int_{\partial \mathbb{D}} \|g\|_\infty^2 \omega_\lambda^2(f) |d\lambda| \\ &\lesssim \|fg\|_{\mathcal{D}_K(\partial \mathbb{D})}^2 + \|g\|_\infty^2 \|f\|_{\mathcal{D}_K(\partial \mathbb{D})}^2 \\ &\lesssim \|f\|_{\mathcal{D}_K(\partial \mathbb{D})}^2. \end{aligned}$$

For $f \in \mathcal{D}_K$, we have $|\nabla \widehat{f}(z)| \approx |f'(z)|$. Theorem 4.1 gives

$$\int_{\mathbb{D}} |f(z)|^2 |\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2.$$

Thus, $|\nabla \widehat{g}(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. $\square$

Remark 4.7 Let $f \in L^\infty(\partial \mathbb{D})$ and $F \in C^1(\mathbb{D}) \cap L^\infty(\mathbb{D})$ with $\lim_{r \rightarrow 1} F(re^{it}) = f(e^{it})$ a.e. $e^{it} \in \partial \mathbb{D}$. From the proof of Theorem 4.6, if $|\nabla F(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, then $f \in M(\mathcal{D}_K(\partial \mathbb{D}))$.

If K satisfies the conditions in Theorem D, Kerman and Sawyer [21] proved that $f \in M(\mathcal{D}_K)$ if and only if $f \in H^\infty$ and $|f'(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. As a consequence of Theorem 4.6, we obtain the following relationship between $M(\mathcal{D}_K)$ and $M(\mathcal{D}_K(\partial \mathbb{D}))$.

Corollary 4.8 Let K satisfy (1.1) and (1.2), and $f \in H^2$. Then $f \in M(\mathcal{D}_K)$ if and only if $f_b \in M(\mathcal{D}_K(\partial \mathbb{D}))$.

Lemma F [31] Let K satisfy (1.1) and (1.2). Then

$$\int_{\mathbb{D}} |f(z) - f(0)|^2 K(1 - |z|^2) dA(z) \lesssim \int_{\mathbb{D}} (1 - |z|^2)^2 |f'(z)|^2 K(1 - |z|^2) dA(z)$$

for all $f \in H(\mathbb{D})$.

Lemma 4.9 Let K satisfy (1.1) and (1.2). Then $e^{-i\theta} \in M(\mathcal{D}_K(\partial \mathbb{D}))$, $\theta \in [0, 2\pi]$.

Proof Clearly, $e^{-i\theta} \in M(\mathcal{D}_K(\partial \mathbb{D}))$ if and only if $e^{i\theta} \in M(\mathcal{D}_K(\partial \mathbb{D}))$. Note that $\widehat{e^{i\theta}} = z$. By Theorem 4.6, it is sufficient to show that $K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. Let $f \in \mathcal{D}_K$. Using Lemma F, we have

$$\begin{aligned} \int_{\mathbb{D}} |f(z)|^2 K(1 - |z|^2) dA(z) &\lesssim |f(0)|^2 + \int_{\mathbb{D}} |f(z) - f(0)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim |f(0)|^2 + \int_{\mathbb{D}} (1 - |z|^2)^2 |f'(z)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim \|f\|_{\mathcal{D}_K}^2. \end{aligned}$$

The proof is complete. $\square$

As another main result of the section, the following theorem gives a solution of $\bar{\partial}$ -equation in $\mathcal{D}_K(\partial\mathbb{D})$ spaces.

Theorem 4.10 *Let K satisfy (1.1) and (1.2). If $\phi \in C(\overline{\mathbb{D}})$, the space of functions continuous on $\overline{\mathbb{D}}$, then there exists a function $f \in C(\overline{\mathbb{D}})$ such that*

$$\frac{\partial f}{\partial \bar{z}} = \phi$$

on $\mathbb{D}$ and $f|_{\partial\mathbb{D}} \in \mathcal{D}_K(\partial\mathbb{D})$.

Proof By [17, Chapter VIII], the function

$$f(z) = \int_{\mathbb{D}} \frac{\phi(w)}{z-w} dA(w) \in C(\overline{\mathbb{D}})$$

and $\bar{\partial}f = \phi$. In order to show that $f|_{\partial\mathbb{D}} \in \mathcal{D}_K(\partial\mathbb{D})$, consider the function

$$g(z) = \int_{\mathbb{D}} \frac{\phi(w)}{1-\bar{z}w} dA(w).$$

Then, $g(e^{i\theta}) = e^{i\theta} f(e^{i\theta})$. From Theorem 4.1 and Lemma 4.9, it is sufficient to show that

$$\int_{\mathbb{D}} |\nabla g(z)|^2 K(1-|z|^2) dA(z) < \infty.$$

The proof of Theorem 2.4 in [32] yields

$$\int_{\mathbb{D}} \left(\int_{\mathbb{D}} \frac{|\phi(w)|}{|1-\bar{w}z|^2} dA(w) \right)^2 K(1-|z|) dA(z) \lesssim \int_{\mathbb{D}} |\phi(z)|^2 K(1-|z|) dA(z).$$

Combining this with

$$|\nabla g(z)| \lesssim \int_{\mathbb{D}} \frac{|\phi(w)|}{|1-\bar{w}z|^2} dA(w),$$

we get

$$\int_{\mathbb{D}} |\nabla g(z)|^2 K(1-|z|^2) dA(z) < \infty.$$

The result is proved. $\square$

5 $\bar{\partial}$ -Equation and Corona Theorem for $M(\mathcal{D}_K)$

In this section, using a derivative-free characterization of $\mathcal{D}_K$ spaces and standard techniques, we establish the $\bar{\partial}$ -equation and corona theorem for $M(\mathcal{D}_K)$. Corona type decomposition theorem on $M(\mathcal{D}_K)$ is also obtained.

Lemma 5.1 *Let K satisfy (1.2). If g is a function on $\mathbb{D}$ such that $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, then $|g(z)| dA(z)$ is a Carleson measure.*

Proof For any $I \subset \partial\mathbb{D}$. Note that $\frac{K_2(t)}{t^{1-c}}$ is decreasing. By Hölder's inequality and Theorem D,

$$\begin{aligned} \int_{S(I)} |g(z)| dA(z) &\lesssim \left(\int_{S(I)} |g(z)|^2 K(1 - |z|^2) dA(z) \right)^{\frac{1}{2}} \\ &\quad \times \left(\int_{S(I)} K^{-1}(1 - |z|^2) dA(z) \right)^{\frac{1}{2}} \\ &\lesssim |I|^{\frac{1-c}{2}} \left(\int_{S(I)} (1 - |z|^2)^{c-1} dA(z) \right)^{\frac{1}{2}} \\ &\lesssim |I|. \end{aligned}$$

□

The following result is a derivative-free characterization of $\mathcal{D}_K$ spaces. See [9] for a similar description of Besov type spaces.

Lemma 5.2 *Let K satisfy (1.1) and (1.2), $\tau > -1$ and $\sigma > -1$. Then $f \in \mathcal{D}_K$ if and only if*

$$\int_{\mathbb{D}} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4+\tau+\sigma}} (1 - |z|^2)^\tau (1 - |w|^2)^\sigma K(1 - |w|^2) dA(z) dA(w) < \infty.$$

Proof From the proof of Theorem 1 in [26], it is sufficient to prove the result for the case $\sigma = \tau$. Note that for any $g \in H(\mathbb{D})$,

$$\int_{\mathbb{D}} |g(u) - g(0)|^2 (1 - |u|^2)^\tau dA(u) \approx \int_{\mathbb{D}} |g'(u)|^2 (1 - |u|^2)^{2+\tau} dA(u).$$

By the change of variable $z = \sigma_w(u)$, we obtain

$$\begin{aligned} &\int_{\mathbb{D}} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4+2\tau}} (1 - |z|^2)^\tau (1 - |w|^2)^\tau K(1 - |w|^2) dA(z) dA(w) \\ &= \int_{\mathbb{D}} \int_{\mathbb{D}} \frac{|f(\sigma_w(u)) - f(\sigma_w(0))|^2}{|1 - \bar{w}\sigma_w(u)|^{4+2\tau}} (1 - |\sigma_w(u)|^2)^\tau \frac{(1 - |w|^2)^2}{|1 - \bar{w}u|^4} dA(u) \\ &\quad \times (1 - |w|^2)^\tau K(1 - |w|^2) dA(w) \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{D}} \int_{\mathbb{D}} |f(\sigma_w(u)) - f(\sigma_w(0))|^2 (1 - |u|^2)^\tau dA(u) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w) \\
&\approx \int_{\mathbb{D}} \int_{\mathbb{D}} |(f \circ \sigma_w)'(u)|^2 (1 - |u|^2)^{2+\tau} dA(u) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w).
\end{aligned}$$

Note that $\tau > -1$. Applying Lemma A and the change of variable $u = \sigma_w(z)$ again gives

$$\begin{aligned}
&\int_{\mathbb{D}} \int_{\mathbb{D}} |(f \circ \sigma_w)'(u)|^2 (1 - |u|^2)^{2+\tau} dA(u) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w) \\
&= \int_{\mathbb{D}} \int_{\mathbb{D}} |f'(z)|^2 (1 - |\sigma_w(z)|^2)^{2+\tau} dA(z) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w) \\
&= \int_{\mathbb{D}} |f'(z)|^2 (1 - |z|^2)^{2+\tau} dA(z) \int_{\mathbb{D}} \frac{(1 - |w|^2)^\tau K(1 - |w|^2)}{|1 - \bar{w}z|^{4+2\tau}} dA(w) \\
&\lesssim \int_{\mathbb{D}} |f'(z)|^2 K(1 - |z|^2) dA(z).
\end{aligned}$$

On the other hand, for $0 < r < 1$, we write $E(w, r) = \{z \in \mathbb{D} : |\sigma_w(z)| < r\}$. By the sub-mean-value property of $|f|^2$, we deduce that

$$\begin{aligned}
&\int_{\mathbb{D}} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4+2\tau}} (1 - |z|^2)^\tau (1 - |w|^2)^\tau K(1 - |w|^2) dA(z) dA(w) \\
&\approx \int_{\mathbb{D}} \int_{\mathbb{D}} |f'(z)|^2 (1 - |\sigma_w(z)|^2)^{2+\tau} dA(z) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w) \\
&\gtrsim \int_{\mathbb{D}} \int_{E(w,r)} |f'(z)|^2 (1 - |\sigma_w(z)|^2)^{2+\tau} dA(z) \frac{K(1 - |w|^2)}{(1 - |w|^2)^2} dA(w) \\
&\approx \int_{\mathbb{D}} \frac{1}{(1 - |w|^2)^2} \int_{E(w,r)} |f'(z)|^2 dA(z) K(1 - |w|^2) dA(w) \\
&\gtrsim \int_{\mathbb{D}} |f'(w)|^2 K(1 - |w|^2) dA(w).
\end{aligned}$$

The proof is complete. $\square$

Lemma 5.3 *Let K satisfy (1.1) and (1.2). If g is a function on $\mathbb{D}$ such that $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, then $|T_g(z)|^2 K(1 - |z|^2) dA(z)$ is also a Carleson measure for $\mathcal{D}_K$. Here*

$$T_g(z) = \int_{\mathbb{D}} \frac{|g(w)|}{|1 - \bar{w}z|^2} dA(w).$$

Proof If $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, that is

$$\int_{\mathbb{D}} |f(z)|^2 |g(z)|^2 K(1 - |z|^2) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2$$

for all $f \in \mathcal{D}_K$. In particular, take $f_w(z) = (1 - \bar{w}z)^{-\alpha/2}$, $w \in \mathbb{D}$, $\alpha > 0$. We get

$$\int_{\mathbb{D}} \frac{|g(z)|^2 K(1 - |z|^2)}{|1 - \bar{w}z|^\alpha} dA(z) \lesssim \int_{\mathbb{D}} \frac{K(1 - |z|^2)}{|1 - \bar{w}z|^{2+\alpha}} dA(z).$$

From Lemma 2.1 in [33], there exists $\varepsilon \in (0, 1)$ such that

$$\int_{\mathbb{D}} \frac{K(1 - |w|^2)}{|1 - \bar{w}z|^{2+\varepsilon}} dA(w) \lesssim \frac{K(1 - |z|^2)}{(1 - |z|^2)^\varepsilon}.$$

Thus,

$$\begin{aligned} |(f|T_g - T_{fg})(z)|^2 &\leq \left(\int_{\mathbb{D}} \frac{|g(w)||f(z) - f(w)|}{|1 - \bar{w}z|^2} dA(w) \right)^2 \\ &\lesssim \int_{\mathbb{D}} \frac{|g(w)|^2 K(1 - |w|^2)}{|1 - \bar{w}z|^\varepsilon} dA(w) \\ &\quad \times \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \frac{dA(w)}{K(1 - |w|^2)} \\ &\lesssim \int_{\mathbb{D}} \frac{K(1 - |w|^2)}{|1 - \bar{w}z|^{2+\varepsilon}} dA(w) \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \frac{dA(w)}{K(1 - |w|^2)} \\ &\lesssim \frac{K(1 - |z|^2)}{(1 - |z|^2)^\varepsilon} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \frac{dA(w)}{K(1 - |w|^2)} \\ &\lesssim \frac{1}{(1 - |z|^2)^\varepsilon} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \frac{K(|1 - \bar{w}z|)}{K(1 - |w|)} dA(w) \\ &\lesssim \frac{1}{(1 - |z|^2)^\varepsilon} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \varphi_K \left(\frac{|1 - \bar{w}z|}{1 - |w|} \right) dA(w) \\ &\lesssim \frac{1}{(1 - |z|^2)^\varepsilon} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{4-\varepsilon}} \left(\frac{|1 - \bar{w}z|}{1 - |w|} \right)^{1-c} dA(w) \\ &\approx \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{3+c-\varepsilon}} (1 - |w|^2)^{c-1} (1 - |z|^2)^{-\varepsilon} dA(w). \end{aligned}$$

Combining this with Lemma 5.2 gives

$$\begin{aligned} &\int_{\mathbb{D}} |(f|T_g - T_{fg})(z)|^2 K(1 - |z|^2) dA(z) \\ &\lesssim \int_{\mathbb{D}} \int_{\mathbb{D}} \frac{|f(z) - f(w)|^2}{|1 - \bar{w}z|^{3+c-\varepsilon}} (1 - |w|^2)^{c-1} (1 - |z|^2)^{-\varepsilon} K(1 - |z|^2) dA(w) dA(z) \\ &\lesssim \|f\|_{\mathcal{D}_K}^2. \end{aligned}$$

From the proof of Theorem 2.4 in [32], we get that for any $I \subset \partial\mathbb{D}$,

$$\begin{aligned} & \int_{S(I)} \left(\int_{\mathbb{D}} \frac{|f(w)g(w)|}{|1 - \bar{w}z|^2} dA(w) \right)^2 K \left(\frac{1 - |z|}{|I|} \right) dA(z) \\ & \lesssim \int_{S(2I)} |f(z)g(z)|^2 K \left(\frac{1 - |z|}{|I|} \right) dA(z). \end{aligned}$$

Taking $|I| = 1$ in the proof of Theorem 2.4 in [32], we get

$$\int_{\mathbb{D}} |T_{fg}(z)|^2 K(1 - |z|) dA(z) \lesssim \int_{\mathbb{D}} |f(z)g(z)|^2 K(1 - |z|) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2.$$

Thus,

$$\int_{\mathbb{D}} |f(z)T_g(z)|^2 K(1 - |z|) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2.$$

□

Hörmander [19] proved that if $|g(z)|dA(z)$ is a Carleson measure, then there exists a function f defined on $\mathbb{D} \cup \partial\mathbb{D}$ such that $\partial f / \partial \bar{z} = g(z)$ on $\mathbb{D}$, and such that the boundary value function f_b is in $L^\infty(\partial\mathbb{D})$. Jones [20] gave a constructive method to solve the $\bar{\partial}$ -equation.

Theorem 5.4 *Let K satisfy (1.1) and (1.2). If g is a function on $\mathbb{D}$ such that $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, then there exists a function f defined on $\mathbb{D} \cup \partial\mathbb{D}$ such that*

$$\frac{\partial f}{\partial \bar{z}} = g$$

on $\mathbb{D}$ and $f|_{\partial\mathbb{D}} \in M(\mathcal{D}_K(\partial\mathbb{D}))$.

Proof If $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, Lemma 5.1 gives that $d\mu(z) = |g(z)|dA(z)$ is a Carleson measure. We take P. Jones' constructive solution u , of the $\bar{\partial}$ -problem $\bar{\partial}u = g$ in [20]. Then u has radial limits $u|_{\partial\mathbb{D}} \in L^\infty(\partial\mathbb{D})$. Moreover, the function

$$v(z) = \frac{2i}{\pi} \int_{\mathbb{D}} \frac{1 - |\xi|^2}{|1 - \bar{\xi}z|^2} \exp \left(\int_{|w| \geq |\xi|} \left(\frac{1 + \bar{w}\xi}{1 - \bar{w}\xi} - \frac{1 + \bar{w}z}{1 - \bar{w}z} \right) |g(w)|dA(w) \right) g(\xi) dA(\xi)$$

has the same boundary values as $zu(z)$. Then, by Remark 4.7 and Lemma 4.9, it is sufficient to show that $v \in L^\infty(\mathbb{D})$ and $|\nabla v(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$.

By the proof of Lemma 2.1 in [20], we have

$$|\nabla v(z)| \lesssim \int_{\mathbb{D}} \frac{|g(w)|}{|1 - \bar{w}z|^2} dA(w).$$

Since $|g(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$, by Lemma 5.3, $|\nabla v(z)|^2 K(1 - |z|^2) dA(z)$ is also a Carleson measure for $\mathcal{D}_K$. Note that $|g(z)| dA(z)$ is a Carleson measure. Then $v \in L^\infty(\mathbb{D})$ by [3, 20] or [35]. $\square$

The following result establishes the corona theorem for $M(\mathcal{D}_K)$.

Theorem 5.5 *Let K satisfy (1.1) and (1.2). If $g_1, \dots, g_n \in M(\mathcal{D}_K)$ with*

$$\inf_{z \in \mathbb{D}} \sum_{j=1}^n |g_j(z)| =: \delta > 0,$$

then there exist functions $f_1, \dots, f_n \in M(\mathcal{D}_K)$ such that

$$f_1 g_1 + \dots + f_n g_n \equiv 1.$$

Proof Consider the non-analytic solutions

$$\varphi_j = \frac{\overline{g_j}}{\sum_{k=1}^n |g_k|^2}, \quad j = 1, \dots, n,$$

of the equation $g_1 \varphi_1 + \dots + g_n \varphi_n = 1$. For $1 \leq j, k \leq n$, let $G_{j,k} = \varphi_j \bar{\partial} \varphi_k$. Then

$$|G_{j,k}(z)| \lesssim \delta^{-3} \sum_{i=1}^n |g'_i(z)|.$$

Since $g_i \in M(\mathcal{D}_K)$, $1 \leq i \leq n$, we get that $|g'_i(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. Hence,

$$\begin{aligned} & \int_{\mathbb{D}} |f(z)|^2 |G_{j,k}(z)|^2 K(1 - |z|^2) dA(z) \\ & \lesssim \sum_{i=1}^n \int_{\mathbb{D}} |f(z)|^2 |g'_i(z)|^2 K(1 - |z|^2) dA(z) \lesssim \|f\|_{\mathcal{D}_K}^2, \end{aligned}$$

for all $f \in \mathcal{D}_K$. Thus, $|G_{j,k}(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. By Theorem 5.4, there exist functions a_{jk} with boundary values in $\mathcal{D}_K(\partial\mathbb{D})$ such that $\bar{\partial} a_{jk} = G_{j,k}$, for $1 \leq j, k \leq n$. Then functions

$$f_j = \varphi_j + \sum_{k=1}^n (a_{jk} - a_{kj}) g_k, \quad j = 1, \dots, n,$$

are analytic solutions of the equation $\sum_{j=1}^n f_j g_j = 1$.

To prove $f_j \in M(\mathcal{D}_K)$, by Corollary 4.8, it is sufficient to show that f_j has boundary value in $M(\mathcal{D}_K(\partial\mathbb{D}))$. Since $g_j \in M(\mathcal{D}_K)$, and a_{jk} has boundary value in $M(\mathcal{D}_K(\partial\mathbb{D}))$, we need only to show that the boundary value of φ_j is in $M(\mathcal{D}_K(\partial\mathbb{D}))$.

By Remark 4.7, it suffices to show $\varphi_j \in L^\infty(\mathbb{D})$ and $|\nabla\varphi_j(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. Since $g_j \in M(\mathcal{D}_K) \subseteq H^\infty$, we get that $\varphi_j \in L^\infty(\mathbb{D})$. A direct computation gives

$$|\nabla\varphi_j(z)| \lesssim \delta^{-3} \sum_{i=1}^n |g_i'(z)|.$$

It follows that $|\nabla\varphi_j(z)|^2 K(1 - |z|^2) dA(z)$ is a Carleson measure for $\mathcal{D}_K$. The proof is complete. $\square$

For $g = (g_1, \dots, g_n)$, $g_i \in H(\mathbb{D})$, and $f = (f_1, \dots, f_n)$, $f_i \in H(\mathbb{D})$, $i = 1, 2, \dots, n$, define the operator S_g as

$$S_g(f) = \sum_{k=1}^n f_k g_k.$$

We have the following corona type decomposition of $M(\mathcal{D}_K)$.

Corollary 5.6 *Let K satisfy (1.1) and (1.2). The following conditions are equivalent.*

- (1) S_g maps $M(\mathcal{D}_K) \times \dots \times M(\mathcal{D}_K)$ boundedly onto $M(\mathcal{D}_K)$.
- (2) $g_1, \dots, g_n \in M(\mathcal{D}_K)$ and satisfy

$$\inf_{z \in \mathbb{D}} \sum_{j=1}^n |g_j(z)| =: \delta > 0.$$

Proof (1) $\Rightarrow$ (2). If S_g maps $M(\mathcal{D}_K) \times \dots \times M(\mathcal{D}_K)$ boundedly onto $M(\mathcal{D}_K)$, then the functions $g_j \in M(\mathcal{D}_K)$, $j = 1, 2, \dots, n$. Let $f \in M(\mathcal{D}_K)$. Then there exist functions $f_j \in M(\mathcal{D}_K)$ with $f = \sum_{j=1}^n f_j g_j$. Moreover, by the open mapping theorem, $\|f_j\|_{M(\mathcal{D}_K)} \lesssim \|f\|_{M(\mathcal{D}_K)}$. Since $M(\mathcal{D}_K) \subset H^\infty$, we have $|f_j(z)| \lesssim \|f_j\|_{M(\mathcal{D}_K)}$ for all $z \in \mathbb{D}$. Taking $f = 1$, we obtain that

$$1 = |f(z)| \leq \sum_{j=1}^n |f_j(z)| |g_j(z)| \lesssim \|f\|_{M(\mathcal{D}_K)} \sum_{j=1}^n |g_j(z)|,$$

which gives the desired result.

(2) $\Rightarrow$ (1). The boundedness of S_g is obvious. By Theorem 5.5, there exist functions $h_1, \dots, h_n \in M(\mathcal{D}_K)$ with $\sum_{k=1}^n h_k g_k = 1$. Then, $f \in M(\mathcal{D}_K)$ implies that the functions $f_k = f h_k$ are in $M(\mathcal{D}_K)$. Thus, S_g is surjective. The proof is complete. $\square$

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